Math Retresher for Masters

Homework 4.

1 $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

□ 1) $n=1$: $\sum_{k=1}^{1} k^2 = 1 = \frac{1\cdot(1+1)(2\cdot1+1)}{6} = \frac{2\cdot3}{6} = 1$

2) let's $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$

3) $\sum_{k=1}^{n+1} k^2 = \sum_{k=1}^{n} k^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + \frac{6(n^2+2n+1)}{6} =$

$= \frac{2n^3 + n^2 + 2n^2 + n + 6n^2 + 12n + 6}{6} = \frac{2n^3 + 9n^2 + 13n + 6}{6} =$

$= \frac{(n+1)(2n^2+7n+6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}$ ■

2 $n^3 \leq 2^{n+1}$, $n \geq 8$

□ 1) $n=8$: $8^3 = (2^3)^3 = 2^9 = 512$, $2^{8+1} = 2^9 = 512 \Rightarrow 8^3 \leq 2^{8+1}$

2) Let $n^3 \leq 2^{n+1}$

3) $(n+1)^3 = n^3 + 3n^2 + 3n + 1$

$2^{n+1+1} = 2\cdot 2^{n+1} \geq 2n^3 = n^3 + n^3$

$(n^3 + 3n^2 + 3n + 1) \vee (n^3 + n^3) \quad | -n^3$

$3n^2 + 3n + 1 \vee n^3$

$0 \vee n^3 - 3n^2 - 3n - 1 = (n-1)^3 - 6n$

We must prove that $6n \leq (n-1)^3$, $n \geq 8$

a) $6 \cdot 8 \;\vee\; (8-1)^3$

$48 \;\vee\; 49 \cdot 7 \Rightarrow$ done

b) Let $6n \leq (n-1)^3$

c) $6(n+1) \;\vee\; (n+1-1)^3$

$6n+6 \;\vee\; n^3$

$6n+6 \leq (n-1)^3+6 \;\vee\; n^3$

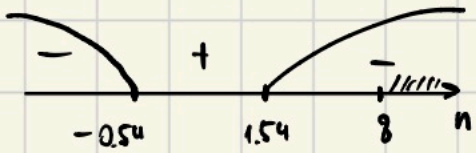
$n^3 - 3n^2 + 3n + 5 \;\vee\; n^3$

We need $-3n^2+3n+5 < 0$

$3n^2 - 3n - 5 > 0$

$D = 9 + 2 \cdot 3 \cdot 5 = 39$

$n_{1,2} = \dfrac{3 \pm \sqrt{39}}{6}$



$\Rightarrow$ this is true for $n \geq 8$

$\Rightarrow (n+1)^3 \leq 2^{n+2}$, $n \geq 8$ ■

3 The sequence X_n does not converge to 0.

Converge : $\forall \varepsilon > 0 \;\; \exists N \in \mathbb{N} \;\; \forall n > N : |X_n| < \varepsilon$

$X_n \not\to 0 \iff \boxed{\exists \varepsilon > 0 \;\; \forall N \in \mathbb{N} \;\; \exists n > N : |X_n| \geq \varepsilon}$

4 $\lim\limits_{n\to\infty} \dfrac{3-2n}{4n+5} = \lim\limits_{n\to\infty} \dfrac{-2+\overset{\to 0}{\frac{3}{n}}}{4+\underset{\to 0}{\frac{5}{n}}} = -\dfrac{2}{4} = \boxed{-\dfrac{1}{2}}$

5 $\lim\limits_{n\to\infty} \dfrac{2n^2 - 3n + 2^n + 5\cdot\sin(n^8) - n!}{4n^{10} - 5n + 2n! + n^2\cdot\sin(n^8) - 10\ln(n)} = \left[\text{the most poverful is } n!\right] =$

$= \lim\limits_{n\to\infty} \dfrac{-n!}{2n!} = \dfrac{-1}{2} = \boxed{-\dfrac{1}{2}}$

6 $\lim\limits_{\substack{n\to\infty \\ n\geq 2}} \left(\sqrt{n^2-n+2} - \sqrt{n^2-n-1}\right) = \lim\limits_{n\to\infty} \dfrac{\left(\sqrt{n^2-n+2} - \sqrt{n^2-n-1}\right)\left(\sqrt{n^2-n+2} + \sqrt{n^2-n-1}\right)}{\sqrt{n^2-n+2} + \sqrt{n^2-n-1}} =$

$$= \lim_{n\to\infty} \frac{(n^2-n+2)-(n^2-n-1)}{\sqrt{n^2-n+2}+\sqrt{n^2-n-1}} = \lim_{n\to\infty} \frac{3}{\sqrt{n^2-n+2}+\sqrt{n^2-n-1}} = 0$$

$\sqrt{n^2-n+2}+\sqrt{n^2-n-1} \to \infty$

7 $X_n = \left(1+\frac{1}{n}\right)^n$

We know $(1+x)^n \geq 1+nx \quad \forall n \geq -1$.

□ 1) Let's show that $X_{n+1} > X_n$. Consider $\frac{X_{n+1}}{X_n}$. Want $\frac{X_{n+1}}{X_n} > 1$

$$\frac{\left(1+\frac{1}{n+1}\right)^{n+1}}{\left(1+\frac{1}{n}\right)^n} = \left(\frac{1+\frac{1}{n+1}}{1+\frac{1}{n}}\right)^n \left(1+\frac{1}{n+1}\right) = \left(\frac{\frac{n+2}{n+1}}{\frac{n+1}{n}}\right)^n \left(1+\frac{1}{n+1}\right) = \left(\frac{n(n+2)}{(n+1)^2}\right)^n \left(1+\frac{1}{n+1}\right) =$$

$$= \left(\frac{(n+1)^2-1}{(n+1)^2}\right)^n \left(1+\frac{1}{n+1}\right) = \left(1-\frac{1}{(n+1)^2}\right)^n \cdot \frac{n+2}{n+1} \geq \left[\text{use Bernoulli's inequality } x=-\frac{1}{(n+1)^2}\right] \geq$$

$$\geq \left(1-\frac{n}{(n+1)^2}\right)\frac{n+2}{n+1} = \frac{n^2+n+1}{(n+1)^2}\cdot\frac{n+2}{n+1} = \frac{(n+2)(n^2+n+1)}{(n+1)^3} = \frac{n^3+3n^2+3n+2}{(n+1)^3} =$$

$$= \frac{(n+1)^3+1}{(n+1)^3} = 1+\frac{1}{(n+1)^3} \geq 1 \Rightarrow \frac{X_{n+1}}{X_n} > 1 \Rightarrow X_{n+1} > X_n$$

2) Take $y_n = \left(1+\frac{1}{n}\right)^{n+1}$, then $y_n > X_n$, because: $y_n = \left(1+\frac{1}{n}\right)^n \cdot \left(1+\frac{1}{n}\right) = X_n\left(1+\frac{1}{n}\right) \geq X_n$

3) Prove that $y_{n+1} < y_n$. Consider $\frac{y_n}{y_{n+1}}$. Want $\frac{y_n}{y_{n+1}} > 1$.

$$\frac{\left(1+\frac{1}{n}\right)^{n+1}}{\left(1+\frac{1}{n+1}\right)^{n+2}} = \left(\frac{1+\frac{1}{n}}{1+\frac{1}{n+1}}\right)^{n+1} \cdot \frac{1}{1+\frac{1}{n+1}} = \left(\frac{\frac{n+1}{n}}{\frac{n+2}{n+1}}\right)^{n+1} \cdot \frac{1}{\frac{n+2}{n+1}} = \left(\frac{(n+1)^2}{n^2+2n}\right)^{n+1} \cdot \frac{n+1}{n+2} =$$

$$= \left(\frac{n^2+2n+1}{n^2+2n}\right)^{n+1} \cdot \frac{n+1}{n+2} = \left(1+\frac{1}{n^2+2n}\right)^{n+1} \cdot \frac{n+1}{n+2} \geq \left[\text{using Bernoulli's inequality } x=\frac{1}{n^2+2n}\right] \geq$$

$$\geq \left(1+\frac{n+1}{n^2+2n}\right)\frac{n+1}{n+2} = \frac{(n^2+3n+1)(n+1)}{n(n+2)^2} = \frac{n^3+4n^2+4n+1}{n(n+2)^2} = \frac{n(n+2)^2+1}{n(n+2)^2} \geq 1 \Rightarrow y_n > y_{n+1}$$

4) $X_{n+1} > X_n$, $y_{n+1} < y_n$, $X_n < y_n \Rightarrow X_n < y_n \leq y_1 = \left(1+\frac{1}{1}\right)^{1+1} = 4 \Rightarrow X_n$ is increasing and bounded above $\Rightarrow$ X_n converges. ■